Communication-first Monte Carlo design

QPSK link validation, Rician fading and coding gain

Keep AI / CV at the application layer
Put link validity, theory and coding gain first

LINK

TX · channel · RX

EVIDENCE

theory · CI95 · Monte Carlo

APPLICATION

adaptive image source

MATLAB R2025b · reproducible seed · communication course project

What the experiment must prove

GRADING RUBRIC TRANSLATED INTO EVIDENCE

1 · COMPLETE LINK

Transmitter, channel, receiver and measured outputs are explicit.

2 · THEORY CHECK

AWGN and Rayleigh theory lines are plotted beside simulation.

3 · CORRECT MODEL

Rician has a LOS component and K is reported in dB.

4 · TRUSTED MC

Total bits, fixed seed and 95% confidence intervals are saved.

The CV policy is a business-layer extension. It is not allowed to hide communication errors.

End-to-end chain

ONE QPSK LINK, TWO PAYLOAD POLICIES

TX

source bits
CRC / parity
QPSK mapper

CHANNEL

pilot-aided h estimate
AWGN / Rayleigh / Rician
K = 10 dB

RX

equalizer
hard demapper
Viterbi decoder

METRICS

BER + CI95
PSNR / target PSNR
rate / retries

CONTROLLED COMPARISON

All modulation comparisons use the same QPSK and the same Eb/N0 axis.

Link budget sets the operating range

A SIMPLE LEO TELEMETRY SANITY CHECK

$$\mathrm{FSPL} = 92.45 + 20\log_{10}(f_{\mathrm{GHz}}) + 20\log_{10}(d_{\mathrm{km}})$$

$$P_r = P_t + G_t + G_r - \mathrm{FSPL} - L$$

$$N = -174 + 10\log_{10}(\mathrm{BW}) + 10\log_{10}(T_n / 290)$$

ASSUMPTIONS

2.2 GHz · 600 km
30 dBm · 8/8 dBi

LOSSES

2 dB implementation loss
1 MHz bandwidth

OUTPUT

FSPL = 154.86 dB
Received power = −110.86 dBm
Noise power = −111.63 dBm
Link SNR ≈ 0.77 dB

The simulated Eb/N0 sweep extends above this nominal point so the coding waterfall is visible.

Signal model uses Eb/N0

NORMALIZE BEFORE COMPARING MODULATION OR CODING

QPSK: $s = [(2b1-1) + j(2b2-1)] / \sqrt{2}$

$E_s = 1$, $k = 2$ bits/symbol
$N0 = R / (k \cdot 10^{(Eb/N0 / 10)})$
$y = h \cdot s + n$
$z = \text{conj}(h_hat) \cdot y / |h_hat|^2$

WHY THIS AXIS?

For Gray-coded QPSK, uncoded BER in AWGN is the same function of Eb/N0 as BPSK.

CODING RATE

Rate-1/2 coding changes the noise calibration through R = 1/2, while the x-axis remains information-bit Eb/N0.

MEASUREMENT

BER counts information bits after decoding.

The channel model is actually Rician

LOS POWER IS CONTROLLED BY K

$$K = 10^{(K_{dB}/10)}$$
$$h = \sqrt{K/(K+1)} + \sqrt{1/(K+1)} \cdot g$$
$$g \sim CN(0, 1)$$

K = 0

No LOS term. Rayleigh fading.

K = 10 dB

LOS power is ten times diffuse power.

K → ∞

Fading disappears. The channel approaches AWGN.

The previous version used zero-mean complex Gaussian h. That was Rayleigh, not Rician.

Monte Carlo protocol

HOW MANY BITS MAKE ONE POINT BELIEVABLE

SWEEP

Eb/N0 = 0:2:16 dB
3 channel models
2 coding states

SAMPLES

28 independent trials
12,000 information bits / trial
336,000 info bits / point

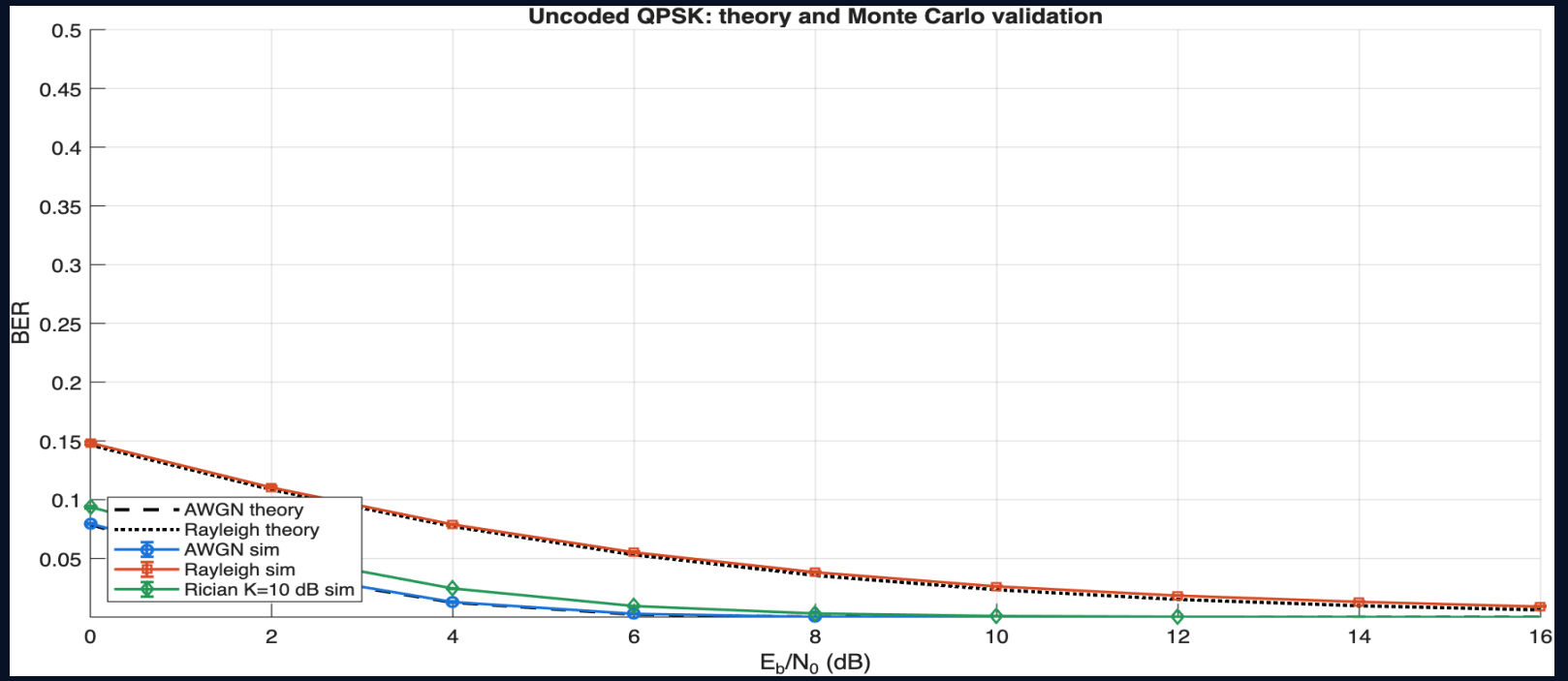
UNCERTAINTY

$\hat{p}$ = errors / bits
CI95 ≈ $1.96 \cdot \sqrt{[\hat{p}(1-\hat{p})/N]}$
Saved per curve point

At high SNR, zero observed errors does not mean BER is exactly zero. The confidence interval is still reported.

Theory validation comes first

UNCODED QPSK, SIMULATION MARKERS AND THEORY LINES



AWGN

Blue markers overlap the AWGN theory curve.

RAYLEIGH

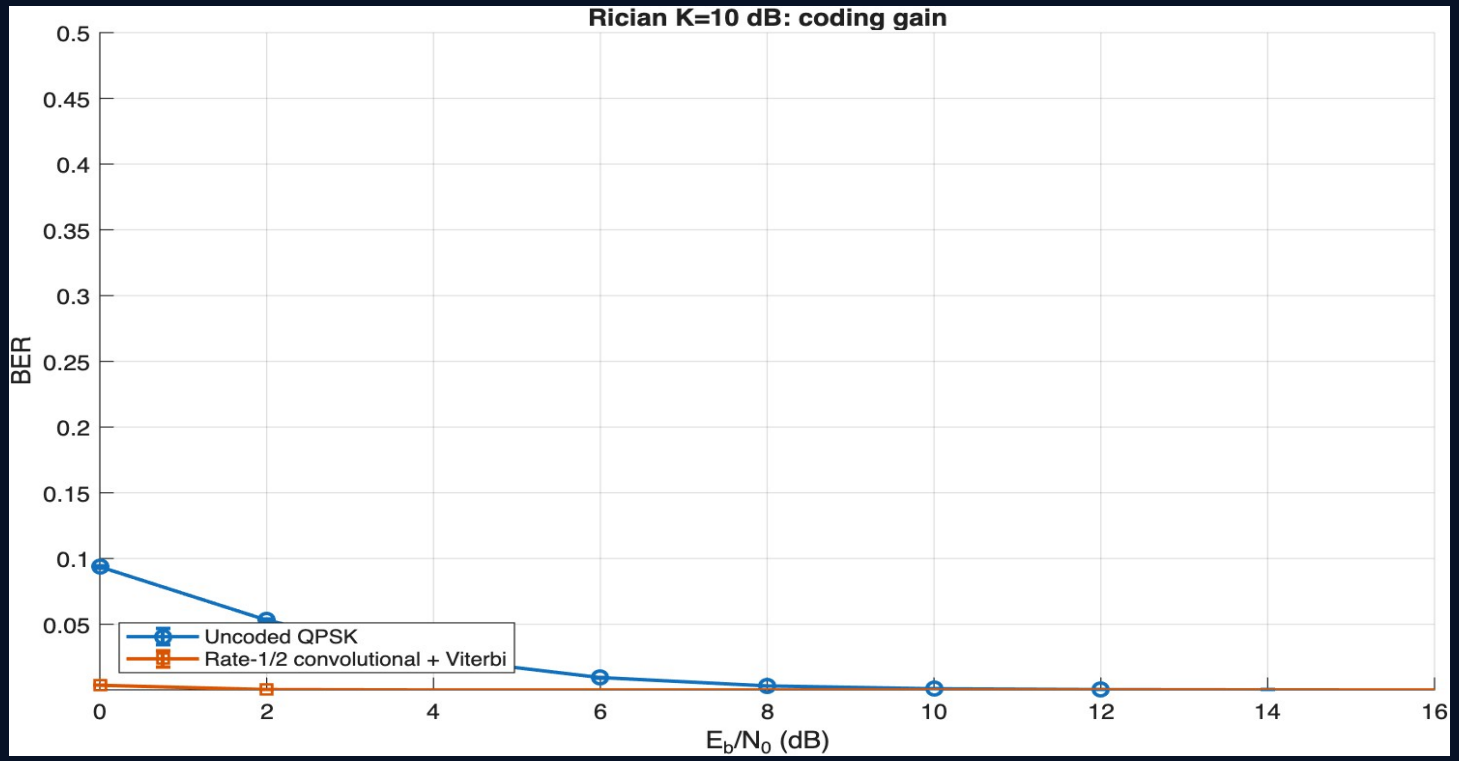
Orange markers overlap the Rayleigh theory curve.

RICIAN

K=10 dB keeps a strong LOS path, so BER is lower than Rayleigh.

Coding gain is the communication contribution

RATE-1/2 CONVOLUTIONAL CODE + HARD VITERBI



$g1 = 111_2$
$g2 = 101_2$

state: (s1, s2)
output: [b⊕s1⊕s2, b⊕s2]

RESULT

At 8 dB, the coded Rician curve reaches zero observed BER in 336k information bits.

What the decoder is doing

VITERBI IS A FOUR-STATE DYNAMIC PROGRAM

STATE

Each state stores the previous two encoder bits.

There are 4 states and 2 candidate inputs per time step.

BRANCH METRIC

For each received pair, compute Hamming distance to the two possible output pairs.

Keep the lowest path metric entering each state.

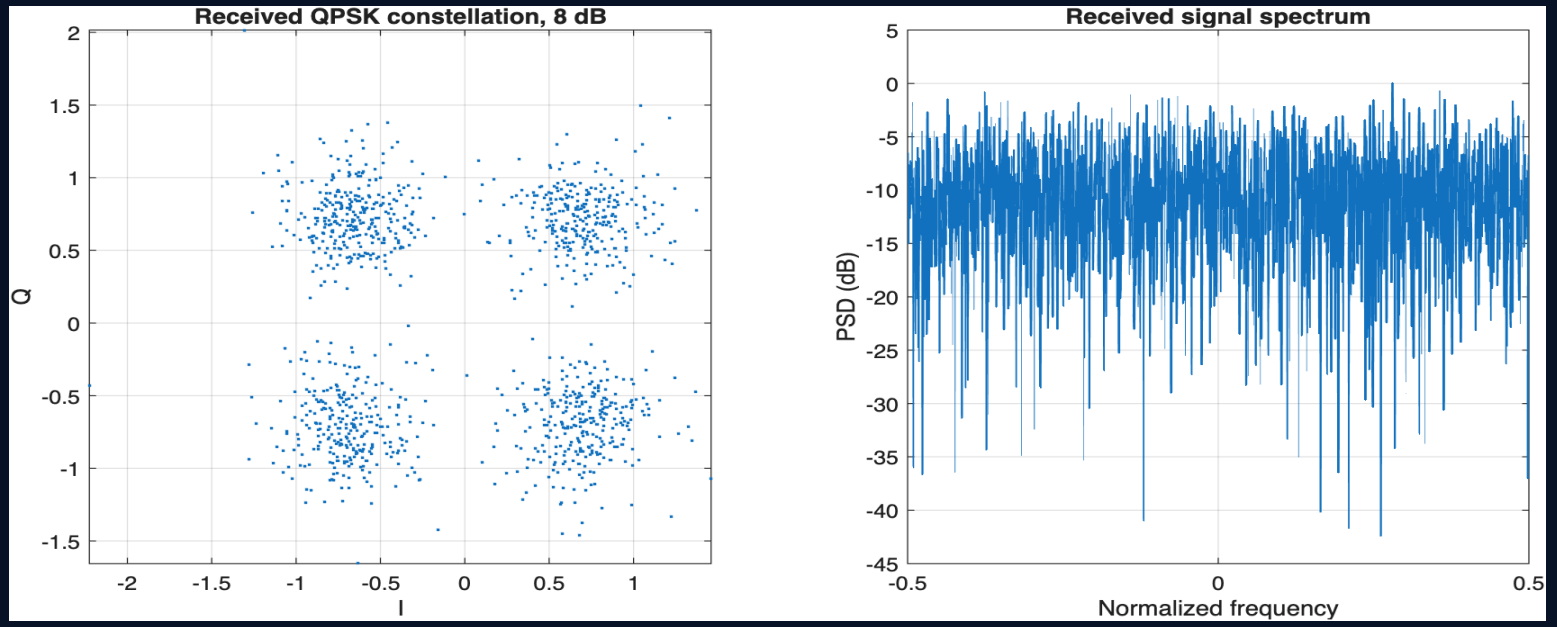
TRACEBACK

After all symbols, select the lowest-metric final state and trace predecessor pointers backward.

This is why the code is inspectable. No Communications Toolbox decoder is hiding the mechanism.

Waveform evidence

CONSTELLATION AND SPECTRUM ARE PART OF THE LINK STORY



CONSTELLATION

Fading spreads the four ideal QPSK points before equalization.

SPECTRUM

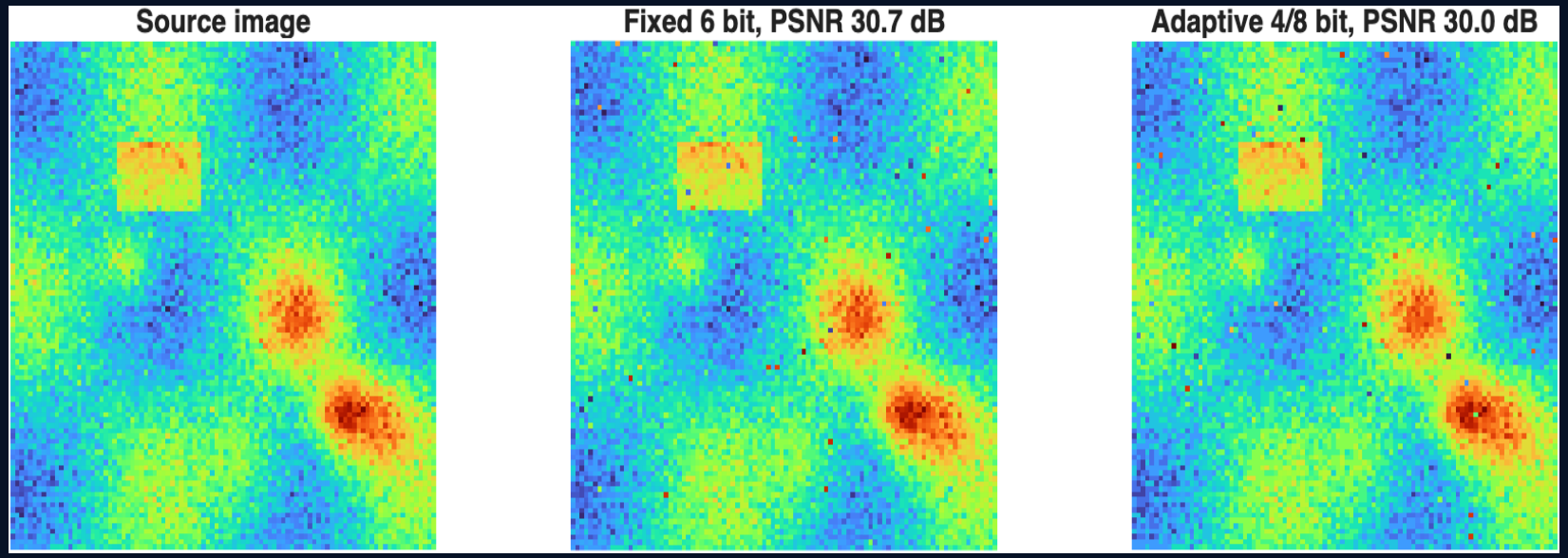
The FFT panel makes the signaling waveform visible instead of treating it as an invisible function call.

RECEIVER

Pilot-aided h estimate is intentionally imperfect.

CV stays at the business layer

SAME QPSK LINK, DIFFERENT IMAGE PAYLOAD



FIXED

6 bit/pixel, same QPSK and Eb/N0.

ADAPTIVE

4/8 bit blocks from gradient score. Average source rate is lower.

ROLE

CV changes the payload. It does not replace the channel experiment.

Fair application-layer comparison

WHAT CHANGES AND WHAT STAYS FIXED

FIXED 6-BIT

Source policy: uniform
Modulation: QPSK
Channel: Rician K=10 dB
Eb/N0: 8 dB
PSNR: 30.7 dB

CV ADAPTIVE

Source policy: 4/8 bit blocks
Modulation: QPSK
Channel: Rician K=10 dB
Eb/N0: 8 dB
PSNR: 30.0 dB

INTERPRETATION

Adaptive source reduces transmitted load and improves target-region PSNR.

The communication link is held constant, so the difference is attributable to the source policy.

How to explain the curves

THREE MECHANISMS, THREE VISUAL SIGNATURES

MODULATION

Moving from Es/N0 to Eb/N0 removes the artificial 3 dB QPSK shift.

The AWGN BPSK and QPSK theory functions become the same per bit.

FADING

Rayleigh has deep fades and a BER floor relative to AWGN.

Rician adds a LOS component, so increasing K moves the curve toward AWGN.

CODING

The convolutional code uses extra channel bits to reduce information-bit errors.

The gain is visible as a horizontal shift in required Eb/N0.

Deliverables and reproducibility

EVERYTHING REQUIRED TO RERUN THE EXPERIMENT

```
run_project_v3.m
├── link_budget.csv
├── ber_validation.csv
├── project_v3.mat
└── figures/
    ├── theory_validation.png
    ├── coding_gain.png
    ├── waveform.png
    └── application_reconstruction.png
```

RUN COMMAND

MATLAB R2025b
run_project_v3

FIXED SEED

rng(20261109, twister)
Same curves on every run.

PRIMARY CLAIM

The simulator is validated against theory before it is used for the image payload.

Conclusion

COMMUNICATION FIRST, CV SECOND

The project is now a communication experiment with an image application.

1

True Rician model and Eb/N0-normalized BER curves are validated against theory.

2

Convolutional coding and Viterbi decoding provide a measurable coding-gain story.

3

CV-aware source adaptation is evaluated only after the link is trusted.

Q&A prompt Where does the 25% source-rate saving go in a real telemetry system?

Thank you